

|                                                         |                                                  |                                                  |                                                         |                                                   |                                           |                                               |
|---------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{K}_{0^+}^{\#1}$                        | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$    |                                                         |                                                   |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger$                     | 0                                                | 0                                                |                                                         |                                                   |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                                | 0                                                | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$               | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$         | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $-\frac{\Upsilon_1 k^2}{2}$                      | $\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}}$ | 0                                                       | 0                                                 |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}}$ | $\frac{\Upsilon_2 k^2}{2}$                       | 0                                                       | 0                                                 |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                                | 0                                                | 0                                                       | 0                                                 |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                                | 0                                                | 0                                                       | 0                                                 | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                                  |                                                  | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{3k^2}{2}\frac{{}^{(4)}\kappa_1}{\sqrt{2}}$ | 0                                         |                                               |
|                                                         |                                                  |                                                  | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                                 | 0                                         |                                               |

|                                                         |                                                          |                                                          |                                                         |                                           |                                           |                                               |
|---------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{T}_{0^+}^{\#1}$                                | $\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$            |                                                         |                                           |                                           |                                               |
| $\mathcal{T}_{0^+}^{\#1}{}^\dagger$                     | 0                                                        | 0                                                        |                                                         |                                           |                                           |                                               |
| $\mathcal{T}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                                        | 0                                                        | $\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$               | $\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$               | $-\frac{k^2 \kappa_4^{(4)}}{2 \sqrt{2} \text{Det}(1^+)}$ | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $-\frac{k^2 \kappa_4^{(4)}}{2 \sqrt{2} \text{Det}(1^+)}$ | $-\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$              | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                                        | 0                                                        | 0                                                       | 0                                         |                                           |                                               |
| $\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                                        | 0                                                        | 0                                                       | 0                                         | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                                          |                                                          | $\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{2}{3 k^2 \kappa_1^{(4)}}$          | 0                                         |                                               |
|                                                         |                                                          |                                                          | $\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                         | 0                                         |                                               |

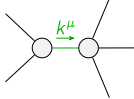
#### Abbreviations used in matrices

$$\Upsilon_1 == 3\frac{{}^{(4)}\kappa_1}{\kappa_1} + 2\frac{{}^{(4)}\kappa_3}{\kappa_3} \&\& \Upsilon_2 == \frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \text{Det}(1^+) == -\frac{1}{8}k^4(6\frac{{}^{(4)}\kappa_1}{\kappa_1}(\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2)$$

#### Lagrangian

$$\frac{{}^{(4)}\kappa_1}{\kappa_1}\partial_\beta\mathcal{K}^\delta_{\chi\delta}\partial^\chi\mathcal{K}^\alpha_{\alpha}{}^\beta + \frac{{}^{(4)}\kappa_3}{\kappa_3}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^\delta + \frac{{}^{(4)}\kappa_4}{\kappa_4}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta -$$

$$3\frac{{}^{(4)}\kappa_1}{\kappa_1}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta - \frac{{}^{(4)}\kappa_3}{\kappa_3}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta + \frac{1}{2}\frac{{}^{(4)}\kappa_3}{\kappa_3}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta + \frac{1}{2}\frac{{}^{(4)}\kappa_4}{\kappa_4}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta$$

|                                                                                    |                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi}\mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Source constraint(s)                                                               | # constraint(s)                                              | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$                                          | 1                                                            | $\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                     | 1                                                            | $\partial_\beta\mathcal{J}^\alpha{}^\beta = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{1^+}^{\#2\alpha} == 0$                                               | 3                                                            | $\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{1^+}^{\#1\alpha} == 0$                                               | 3                                                            | $\partial_\chi\partial^\alpha\mathcal{J}^\beta{}_\chi + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$                                          | 5                                                            | $3\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\mathcal{J}^{\delta\beta\epsilon} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\mathcal{J}^{\delta\alpha}{}_\delta ==$ $3\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\alpha\beta\delta} +$ $2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\mathcal{J}^{\delta\alpha\epsilon} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\mathcal{J}^{\delta\beta}{}_\delta$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Total # constraint(s):                                                             | 13                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Resolved pole(s)                                                                   | # polarization(s)                                            | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Residue                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | 2                                                            | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{24\frac{{}^{(4)}\kappa_1}{\kappa_1}{}^2 + 8\frac{{}^{(4)}\kappa_1}{\kappa_1}(2\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2}{\frac{{}^{(4)}\kappa_1}{\kappa_1}(6\frac{{}^{(4)}\kappa_1}{\kappa_1}(\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4}) + (2\frac{{}^{(4)}\kappa_3}{\kappa_3} + \frac{{}^{(4)}\kappa_4}{\kappa_4})^2)}$ |

#### Resolved unitarity condition(s):

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} < 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{2} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}}) \parallel (-\frac{{}^{(4)}\kappa_4}{2} < \frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}}) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0) \parallel (\frac{{}^{(4)}\kappa_3}{\kappa_3} > -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)))) \parallel$$

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} == 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < 0 \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < -\frac{2\frac{{}^{(4)}\kappa_3}{\kappa_3}}{3}) \parallel (\frac{{}^{(4)}\kappa_3}{\kappa_3} > 0 \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < -\frac{2\frac{{}^{(4)}\kappa_3}{\kappa_3}}{3} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)))) \parallel$$

$$(\frac{{}^{(4)}\kappa_4}{\kappa_4} > 0 \&\& ((\frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& 0 < \frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}}) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0) \parallel (-\frac{{}^{(4)}\kappa_4}{\kappa_4} < \frac{{}^{(4)}\kappa_3}{\kappa_3} < -\frac{{}^{(4)}\kappa_4}{2} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)) \parallel$$

$$(\frac{{}^{(4)}\kappa_3}{\kappa_3} == -\frac{{}^{(4)}\kappa_4}{\kappa_4} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < 0 \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0)) \parallel (\frac{{}^{(4)}\kappa_3}{\kappa_3} > -\frac{{}^{(4)}\kappa_4}{2} \&\& (\frac{{}^{(4)}\kappa_1}{\kappa_1} < \frac{-4\frac{{}^{(4)}\kappa_3}{\kappa_3} - 4\frac{{}^{(4)}\kappa_3}{\kappa_3}\frac{{}^{(4)}\kappa_4}{\kappa_4} - \frac{{}^{(4)}\kappa_4}{\kappa_4}{}^2}{6\frac{{}^{(4)}\kappa_3}{\kappa_3} + 6\frac{{}^{(4)}\kappa_4}{\kappa_4}} \parallel \frac{{}^{(4)}\kappa_1}{\kappa_1} > 0))))$$